ECE448 Spring 2025

Lesson 16

Final Project Intro

UNITED STATES
AIR FORCE
ACADEMY

Final Project Overview

- You pick a project that employs SDRs
- Significant portion of final grade (40%)
- Does not have to use GNURadio
- Culminates in final report and presentation on M40
- Milestones support final report

MILESTONES

1. Proposal Due: M22
2. Background Research: M27
3. Design Description: M31
4. Prototype Update: M35
5. Final Report/presentation due: M40

IDEAS

- FPV Detector/Jammer (I have a starting point)
- Reverse Engineering of radio devices
- ADS-B emitter
- RTL-SDR Blog
- FHSS
- SDR RADAR
- NOAA imagery decoding (antenna design element)

- Non-RF comms (ultrasonic, visual)
- TEMPEST
- LORA

Proposal

M22

- What do you want to do?
- Why do you think it's interesting and reasonable?
- Cursory research review would be helpful
- What is your end goal?
- What is your A-level, B-level, etc. for accomplishment?
- What are your next steps?

Background Research

3-5

- Find some high-quality, relevant papers/blogs/references
- Explain the relationship between the reference and your project
- What will you use from the reference as building blocks?
- What are your next steps?

Background Research

- Find some high-quality, relevant papers/blogs/references
- Summarize their work
- Explain the relationship between the reference and your project
- What are your next steps?

Design Description

- Describe what you want to build
- What are the subsystem/components you will use?
- What are your unanswered questions?
- Where do you expect sticking points?
- What is your testing/verification process?
- What are your next steps?

Prototype Description

- Update on your design/test/build process
- What are your next steps?
- How well are you doing according to your proposal?

Final Report

Code

- Bring it all together
- What did you accomplish?
- What did you not accomplish?
- Confirmation of functionality
- Lessons learned
- Future work